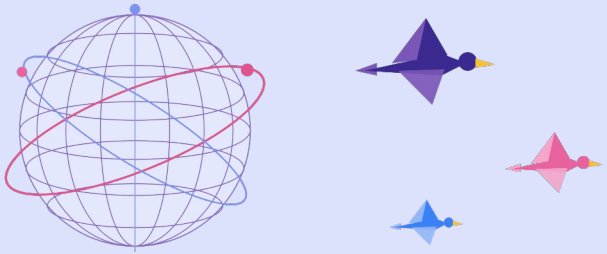
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details



Participant Details

PROJECT TITLE
Quantum-Assisted Traffic Signal Optimization Using QAOA

TRACK
Hackathon

TEAM NAME
NULL POINTER

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Harshit Kumar (Team Lead) · Himansu Rauniyar · Aditya Pattar

INSTITUTION / ORGANISATION
SIT Tumkur · Dept. of CSE

COUNTRY
India

Problem Statement

Fixed timers waste everyone's time

- Most city signals still run fixed timers, blind to the queues in front of them.
- At rush hour, green time goes to empty roads while the busy road backs up.
- Timing N signals over T slots is combinatorial: $2^{N \cdot T}$ possible plans, re-decided every 30 s.

Problem: how do we choose every signal's phase, every 30 s, to minimise waiting across the whole grid, and can QAOA solve it?

83.5 s

average wait per vehicle with fixed timers at rush hour (our 2×2 simulation)

2^{32}

possible plans for a 4×4 grid planned just one minute ahead

Objectives & Novelty

Objectives

1. Model a city signal grid as a QUBO / Ising problem.
2. Solve it with QAOA, built and simulated in Qiskit.
3. Run QAOA as a live controller that re-plans every 30 s.
4. Benchmark quality, cost, scalability and noise vs classical.

What's new

Signal timing, not routing. Earlier quantum traffic work optimises routes; we time the lights.

Green waves emerge. Squared predicted queues couple neighbours; nothing is hand-coded.

QAOA in closed loop. Tested as a live controller on a traffic simulator, deployed online.

Solution & Quantum Advantage

Read queues › Predict › QUBO › QAOA › Apply 30 s

One qubit per signal per slot: $|0\rangle$ = north–south green, $|1\rangle$ = east–west green. Cost = Σ squared queues + switching penalty → Ising $H = \Sigma hZ + \Sigma JZZ$.

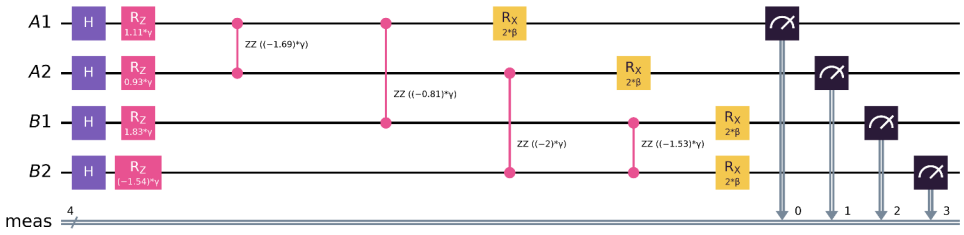
Why quantum

- The QUBO is the native input of QAOA.
- One circuit holds all 2^n plans in superposition.
- Depth p trades circuit cost for quality.

Quantum advantage: honest status

- No speed-up yet: brute force solves 18 qubits in 0.1 s.
- We show feasibility: optimal plans, robust to noise.
- The pipeline is hardware-ready as devices grow.

Algorithm & Circuit Design



$H = \Sigma\, h_k Z_k + \Sigma\, J_{kl} Z_k Z_l \qquad |\gamma,\beta\rangle = \Pi\, e^{-i\beta\Sigma X}\, e^{-i\gamma H}\, |+\rangle^{\otimes n}$

Fig.: one QAOA layer for a 2×2 grid (qubits A1–B2), drawn with qc.draw("mpl").

H superposition of every plan

RZ · RZZ (γ) queue pressure + couplings to neighbours and next slot

RX (β) mixer between plans

Measure 1024 shots, keep the cheapest plan

Trained with COBYLA + INTERP; p = 1–4. At 8 qubits, p = 2: 80 CNOTs, depth 75.

Implementation & Tech Stack

Tech stack

- **Python 3.11** — core language
- **Qiskit 2.x** — circuits · Statevector
- **Qiskit Aer** — noisy shot simulation
- **NumPy · SciPy** — QUBO maths · COBYLA
- **Streamlit** — live dashboard
- **Docker · nginx** — deployed on our VPS

How it is built

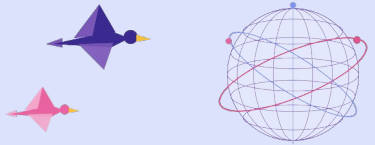
qtraffic/ package: network, simulator, QUBO formulation, QAOA solver, controllers.

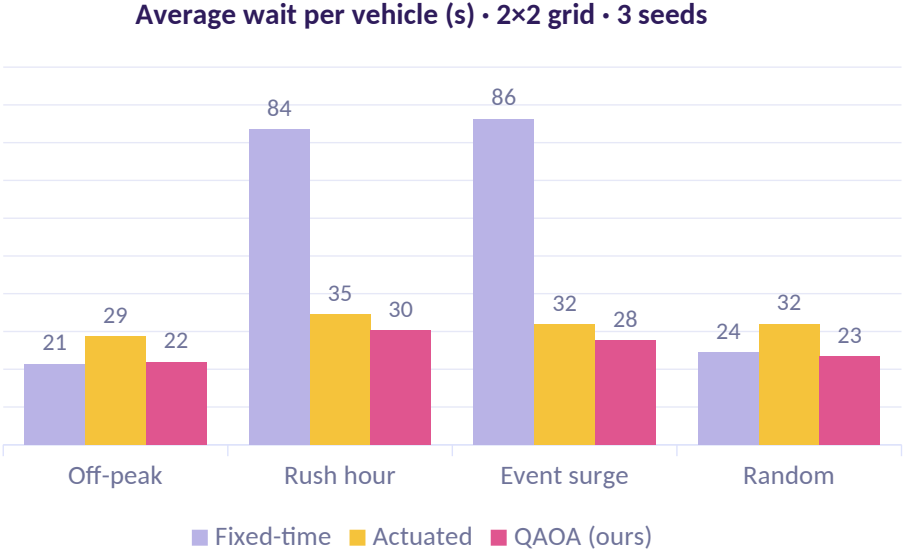
Benchmarks: 4 scenarios × 5 controllers, 3 seeds; up to 32-qubit circuit analysis.

Quality: 26 automated tests, GitHub Actions CI, MIT licence.

Live: qtraffic.harshithk.tech

Results & Analysis





−64%

wait vs fixed timers at rush hour (83.5 → 30.2 s); 12% below actuated.

100% optimal plans found up to 12 qubits

0.85 → 0.93 approx. ratio as p goes 1 → 4

1% noise still optimal on Qiskit Aer

Limit: classical solvers are still faster at this size.

Impact & Real-World Applications

1 Commuters

- Shorter waits at busy junctions.
- Less idling, so less fuel burnt and fewer emissions.
- Smoother green waves along arterials.

2 Cities

- Adaptive control for smart-city corridors.
- Handles event and emergency surges.
- Reuses existing queue detectors.

3 Quantum community

- A real-world QUBO benchmark for QAOA.
- Open source (MIT) and reproducible.
- Ready for IBM Quantum hardware.

Challenges & Future Scope

Challenges we hit

P(optimum) shrinks as qubits grow, so sampling matters.

CNOT count grows with the city: 32 qubits ≈ 450 CNOTs at p = 2.

Noise above 1–2% visibly lowers solution quality.

Aer crashed in containers; fixed by tuning glibc's TLS reserve.

Future scope

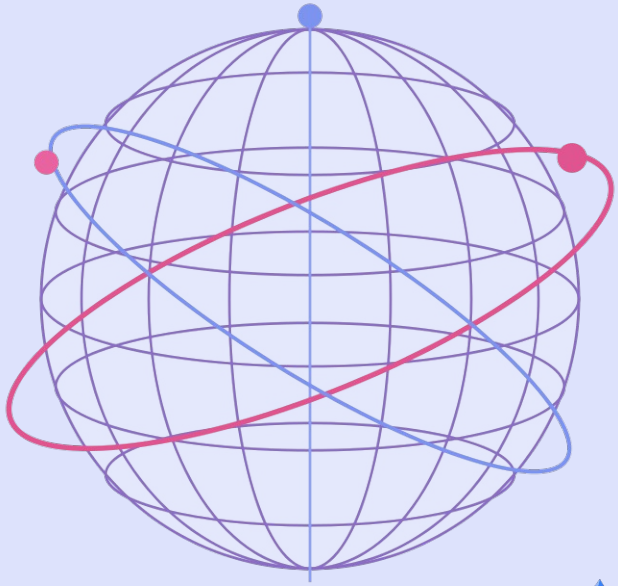
1. **Run on IBM Quantum** via Qiskit Runtime with error mitigation.
2. **Split cities into corridors** that fit on one device.
3. **Real traffic data:** SUMO and live detector feeds.
4. **Richer model:** turning movements and pedestrians.

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



GITHUB REPOSITORY
github.com/Harshithk00/Vhack

DEMO VIDEO LINK
vhack.harshithk.tech

Live dashboard: qtraffic.harshithk.tech

Team NULL POINTER · Harshit Kumar (Lead) · Himansu Rauniyar · Aditya Pattar